ANALYZING DEPRESSION SEVERITY AND CLASSIFICATION

A Major Project Report submitted in partial fulfilment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING
(ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)

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The following is a declaration that the Major Project report, **" ANALYZING DEPRESSION SEVERITY AND CLASSIFICATION "** is an original work completed and submitted to the CSE (AI & ML) Department of Geethanjali College of Engineering & Technology, which is affiliated with Jawaharlal Nehru Technological University Hyderabad. It serves as a record of legitimate project work completed by us under the supervision of **Mr. G Venu Gopal**, Assistant Professor, Department of CSE (AI & ML). We further affirm that none of the work included in this project has been applied for, in whole or in part, towards the award of any other degree or diploma.

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ABSTRACT

Depression is a profound mental health disorder that affects millions globally, disrupting daily routines and overall well-being. Early detection and timely intervention are essential for effectively managing and alleviating its impact. Traditional diagnostic methods, such as the PHQ-9 questionnaire, primarily rely on self-reported symptoms, which may not always provide an accurate assessment due to personal biases or underreporting.

This research introduces a multi-layered approach to depression detection, integrating three key methodologies: PHQ-9 assessment, facial expression analysis, and social media sentiment analysis. The PHQ-9 test serves as a baseline screening tool, offering an initial depression severity score based on user responses. Facial expression analysis leverages deep learning models to detect emotional cues in real time, capturing non-verbal indicators of distress. In addition, natural language processing (NLP) techniques are used to analyze social media comments and text inputs, identifying linguistic patterns and markers that are linked to depression.

Two datasets were utilized for this study: a facial expression dataset containing approximately 35,000 images, and a Twitter dataset comprising 10,000 tweets from individuals exhibiting depressive and non-depressive behavior. By combining these modalities, the system enhances the accuracy and real-time capability of depression detection, overcoming limitations observed in existing single-modality approaches.

The proposed system not only classifies depression severity but also provides personalized motivational feedback and recommendations for psychiatric consultation when necessary. By leveraging advancements in DL and NLP, this project aims to offer an innovative and comprehensive solution to mental health diagnostics, facilitating early detection and timely intervention.

Keywords: Depression Detection, Facial Expression Analysis, Natural Language Processing (NLP), Deep Learning, Social Media Analysis.

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LIST OF ABBREVIATIONS

AI - Artificial Intelligence

ML - Machine Learning

PHQ-9 - Patient Health Questionnaire-9

NLP - Natural Language Processing

CNN - Convolutional Neural Network

FER - Facial Expression Recognition

NLP - Natural Language Processing

EHR - Electronic Health Records

LSTM - Long Short-Term Memory

IEEE - Institute of Electrical and Electronics Engineers

API - Application Programming Interface

DRF - Django REST Framework

E-R Diagram - Entity-Relationship Diagram

1. INTRODUCTION

1.1 Motivation

Depression has emerged as a major public health issue, impacting individuals across all age groups and demographics. The World Health Organization (WHO) reports that more than 280 million people worldwide are affected by depression, making it one of the primary causes of disability. While awareness about mental health has grown, timely diagnosis and intervention remain challenging due to the stigma surrounding mental illnesses and the limitations of conventional diagnostic tools.

Traditional depression assessment methods, such as the PHQ-9 test, rely on self-reported responses, which can be influenced by personal biases, lack of awareness, or unwillingness to disclose mental health struggles. Additionally, clinical evaluations often require professional consultation, which may not be accessible to all individuals. With the increasing use of digital platforms, particularly social media, there is a unique opportunity to leverage technology for more accurate and real-time depression detection.

This project seeks to bridge the gap between traditional mental health assessments and advanced AI-driven diagnostic tools, offering a more innovative and efficient approach to mental health detection. By integrating facial expression analysis, natural language processing (NLP) for text-based sentiment detection, and PHQ-9 scoring, this system provides a holistic and real-time approach to depression detection.

1.2 Problem Statement

Depression is often underdiagnosed due to the limitations of existing detection methods. The PHQ-9 test, while widely used, is based solely on self-reporting, which may not capture subtle emotional distress. Similarly, facial expression analysis and text sentiment analysis, when used independently, may lead to misclassification due to environmental factors, language complexity, or user reluctance to express emotions openly.

Furthermore, existing systems that analyze social media posts for depression indicators often lack contextual understanding, leading to inaccurate predictions. The absence of a

multi-dimensional approach to depression detection results in missed opportunities for timely intervention and support.

This project tackles these challenges by developing a multi-layered system that integrates:

- PHQ-9 questionnaire-based assessment
- Real-time facial expression analysis using DL models
- Social media sentiment analysis using NLP techniques

By integrating these methodologies, the proposed system provides a more precise and dependable framework for detecting depression, enabling individuals to better understand their mental health and access timely support.

1.3 Project Objectives

The primary objectives of this project are:

1. To develop a multi-layered depression detection system that integrates PHQ-9 assessment, facial expression analysis, and social media sentiment analysis.
2. To leverage deep learning and natural language processing techniques for accurate classification of depression severity.
3. To provide real-time feedback and personalized recommendations for individuals at risk of depression.
4. To enhance the accessibility of mental health diagnostics through AI-driven solutions.
5. To ensure data security and privacy in handling sensitive user information.

1.4 Existing System

Existing depression detection systems predominantly rely on:

1. **PHQ-9 Test:** A self-reported questionnaire widely used for depression screening.
2. **Facial Expression Analysis:** Uses image-based emotion detection models to identify signs of depression.
3. **Text-Based Sentiment Analysis:** Processes written content from social media or chat logs to detect depressive tendencies.

While these methods have demonstrated effectiveness, they each have significant limitations when used independently.

1.5 Limitations of Existing Systems

1. **PHQ-9 Self-Reporting Bias:** Users may provide inaccurate responses due to personal biases or reluctance to disclose emotional distress.
2. **Facial Expression Limitations:** Environmental factors such as lighting, background noise, or poor image quality can affect accuracy.
3. **Text-Based Sentiment Misinterpretation:** Social media posts may contain sarcasm, slang, or ambiguous expressions that are difficult to interpret correctly.
4. **Lack of Real-Time Analysis:** Most existing systems do not provide immediate feedback or intervention.

1.6 Proposed System

The proposed system integrates multiple modalities to enhance depression detection accuracy. It combines:

1. **PHQ-9 Test:** Initial depression assessment based on standardized questions.
2. **Facial Expression Analysis:** Detects non-verbal emotional cues in real time.

3. **Social Media Sentiment Analysis:** Analyzes text-based inputs using NLP to identify depressive language patterns.

This integrated approach provides a more holistic understanding of an individual's mental health status, combining various data sources for a thorough assessment.

1.7 Advantages of Proposed System

1. **Multi-Layered Analysis:** Combines multiple assessment methods for higher accuracy.

2. **Real-Time Feedback:** Provides immediate insights based on facial expressions and text analysis.

3. **Personalized Recommendations:** Offers motivational suggestions and professional psychiatric support.

4. **Improved Accuracy:** Reduces bias and environmental limitations through a combined decision-making framework.

5. **Enhanced Accessibility:** Provides a user-friendly interface for individuals to assess their mental health conveniently.

The implementation of this system is anticipated to greatly enhance early detection and intervention strategies for individuals at risk of depression, ultimately contributing to greater mental health awareness and providing valuable support.

2. AIM and OBJECTIVES

2.1 AIM of Proposed System

The main objective of this project is to create an advanced depression detection system that combines multiple analysis techniques, such as the PHQ-9 assessment, facial expression analysis, and social media sentiment analysis. This system aims to deliver real-time, accurate depression classification, facilitating early intervention and support for individuals at risk. By utilizing artificial intelligence, deep learning, and natural language processing (NLP), the proposed system seeks to address the limitations of traditional depression detection methods and enhance mental health diagnostics.

2.2 Objectives of Proposed System

The key objectives of this project are:

1. **To develop a multi-layered depression detection system** that combines PHQ-9 assessment, facial expression analysis, and social media sentiment analysis for accurate and real-time depression classification.

2. **To utilize deep learning models for facial expression recognition**, enabling the system to capture subtle emotional cues associated with depression in real-time through webcam-based analysis.

3. **To implement natural language processing (NLP) techniques** for sentiment analysis of social media text inputs, identifying depressive language patterns and emotional indicators.

4. **To integrate PHQ-9 as a standardized assessment tool**, providing an initial screening mechanism that categorizes users based on their depression severity scores.

5. **To provide real-time feedback and recommendations** based on the combined results of the PHQ-9 test, facial expression analysis, and social media sentiment detection.

6. **To enhance accessibility and usability** by designing an intuitive and user-friendly interface that allows seamless interaction with the system.

7. **To ensure data security and privacy** by implementing encryption and secure data handling practices, protecting user information and complying with ethical standards.

8. **To enable continuous learning and system improvement**, utilizing user feedback and new data inputs to refine model accuracy and reliability over time.

9. **To facilitate professional intervention** by recommending mental health professionals and support services to users identified with moderate to severe depression.

By achieving these objectives, the proposed system aims to revolutionize depression detection, providing a comprehensive, AI-driven solution that enhances mental health diagnostics and early intervention strategies.

3. LITERATURE SURVEY

3.1 Introduction

Several research studies have been conducted in the field of depression detection using various methods, including self-assessment tests, facial expression recognition, and text sentiment analysis. While these approaches have demonstrated effectiveness in identifying depressive symptoms, they often suffer from limitations such as low accuracy, environmental dependencies, and difficulty in capturing real-time emotional states. This section reviews key existing methods and their associated drawbacks.

3.2 Existing Work

1. **Facial Expression Recognition for Depression Detection**
    - **Author(s):** Pantic et al. (2020)
    - **Methodology:** This study explored the use of facial expression recognition (FER) models based on deep learning techniques to detect depression. The model analyzed facial features, including eyebrow movement, lip curvature, and eye openness, to accurately classify emotional states.
    - **Findings:** FER was found to be a reliable indicator of mental health status, providing valuable insights into an individual's emotional well-being.
    - **Drawbacks:** The system's accuracy was significantly affected by environmental factors such as lighting, image resolution, and facial obstructions, leading to inconsistencies in real-world applications.
2. **Social Media as a Predictor for Depression**
    - **Author(s):** De Choudhury et al. (2021)
    - **Methodology:** This research applied NLP techniques to analyze posts for depressive indicators. Sentiment analysis, word frequency, and linguistic markers were used to assess users' mental health status.

- **Findings:** The study demonstrated that social media data could provide early indicators of depression, with significant correlations between negative sentiment and self-reported depressive symptoms.
- **Drawbacks:** The model struggled with sarcasm, ambiguous language, and differences in writing styles, leading to potential misclassifications.

3. **Application of PHQ-9 in Mental Health Screening**
   - **Author(s):** Kroenke et al. (2020)
   - **Methodology:** The PHQ-9 questionnaire was used to screen individuals for depression based on self-reported symptoms.
   - **Findings:** PHQ-9 is widely used in clinical settings and provides a standardized method for diagnosing depression severity.
   - **Drawbacks:** Reliance on self-reported data introduces biases, as individuals may underreport or misinterpret their symptoms.
4. **Deep Learning for Emotion Detection in Text**
   - **Author(s):** Li et al. (2021)
   - **Methodology:** A deep learning-based NLP model was trained on text datasets to classify emotional states, including depression.
   - **Findings:** The model achieved high accuracy in detecting depressive emotions from structured text data.
   - **Drawbacks:** The approach struggled with context-based language variations, such as sarcasm or metaphorical expressions.
5. **Multimodal Depression Detection Using Video and Text**
   - **Author(s):** Singh et al. (2020)

- **Methodology:** This study combined facial expression analysis with text-based sentiment analysis for more accurate depression detection.
- **Findings:** The hybrid approach improved classification accuracy compared to single-method techniques.
- **Drawbacks:** The system required high-quality data input, and its performance degraded in noisy or low-resolution environments.

3.3 Limitations

While existing research has contributed significantly to the field of automated depression detection, several limitations persist:

1. **Single-Modality Constraints:** Many studies rely solely on either text analysis, facial recognition, or self-reported assessments, leading to incomplete evaluations.
2. **Environmental Factors:** Facial recognition models are affected by variations in lighting, facial positioning, and camera quality.
3. **Language Complexity:** NLP-based models face challenges in detecting emotional nuances, sarcasm, and cultural differences in text.
4. **Bias in Self-Reporting:** Tools like PHQ-9 depend on honest self-assessments, which may be influenced by individual perception and stigma.
5. **Lack of Real-Time Analysis:** Most existing models do not provide immediate feedback, reducing their practical applicability in real-world scenarios.

This literature survey emphasizes the importance of a multi-layered approach that combines various depression detection methods to address the limitations of current systems. The proposed research seeks to develop a comprehensive solution that improves accuracy, enables real-time detection, and fosters better user engagement.

Literature Review Comparison:

S.No	Title	Authors	Methods Used	Drawbacks
1	Facial Expression Recognition for Depression Detection	Pantic et al. (2020)	Facial expression analysis using deep learning	Affected by lighting and clarity issues
2	Social Media as a Predictor for Depression	De Choudhury et al. (2021)	NLP and sentiment analysis	Difficulty distinguishing sarcasm and ambiguous expressions
3	Application of PHQ-9 in Mental Health Screening	Kroenke et al. (2020)	PHQ-9 self-report questionnaire	Subject to user bias and underreporting
4	Deep Learning for Emotion Detection in Text	Li et al. (2021)	Deep learning on text data	Struggles with sarcasm and ambiguous language
5	Multimodal Depression Detection Using Video and Text	Singh et al. (2020)	Facial analysis and text sentiment analysis	Requires high-quality input and affected by data clarity and quality

This review and table provide a solid foundation for understanding the current advancements in depression detection, as well as the limitations that the proposed system seeks to address and improve upon.

4. SOFTWARE/ HARDWARE REQUIREMENTS

4.1 Functional Requirements

The functional requirements specify the essential behaviors and features the system must support to ensure its proper operation. These include:

1. **PHQ-9 Integration:** The system should allow users to complete the PHQ-9 depression screening test and compute a depression severity score based on responses.

2. **Facial Expression Detection:** Real-time facial expression analysis should be performed using a webcam, identifying emotions such as sadness, joy, and frustration via deep learning models.

3. **Social Media Sentiment Analysis:** The system should analyze user-provided text (e.g., social media comments) through Natural Language Processing (NLP) to detect depressive sentiments.

4. **Data Integration for Depression Classification:** The system should combine results from PHQ-9, facial analysis, and text sentiment analysis to determine overall depression severity.

5. **Recommendation Engine:** Based on detected depression severity, the system should provide motivational feedback for mild cases and recommend professional help for moderate to severe cases.

6. **User Account Management:** Users should have the ability to create, manage, and delete their accounts, while ensuring that their data is secure and privacy is maintained.

7. **Real-Time Feedback:** The system should deliver immediate responses regarding the user's mental health status.

8. **Data Security:** All sensitive user data should be securely stored and protected in compliance with data privacy standards.

4.2 Non-Functional Requirements

The non-functional requirements outline the key quality attributes that ensure optimal system performance, security, and usability. These include:

1. **Performance:** The system must process input data quickly and deliver real-time analysis without noticeable delays.

2. **Scalability:** The platform should effectively accommodate increasing numbers of users and larger datasets without compromising performance.

3. **Usability:** The user interface must be intuitive and accessible to users with varying levels of technical expertise.

4. **Security:** Data transmission and storage should be encrypted to safeguard sensitive user information from unauthorized access.

5. **Reliability:** The system must ensure high uptime and handle errors smoothly to guarantee consistent and reliable operation.

6. **Cross-Platform Compatibility:** The application should function across web, desktop, and mobile platforms while maintaining a seamless user experience.

4.3 Software Requirements

The software components required for implementation and execution include:

- **Operating System:** Windows 10/11.
- **Programming Language:** Python 3.8+.
- **Web Framework:** Django 5.1.4.
- **Database:** SQLite3.
- **Machine Learning Libraries:** TensorFlow, Keras, Scikit-learn.
- **Natural Language Processing Libraries:** NLTK, SpaCy, Transformers.
- **Frontend Development Tools:** HTML, CSS, JavaScript, Bootstrap.

- **API Integration:** REST API using Django REST Framework (DRF).
- **Version Control:** Git, GitHub.
- **Cloud Deployment (Optional):** Render, AWS, or Heroku.

4.4 Hardware Requirements

The hardware components necessary for smooth execution and real-time processing include:

- **Processor:** Minimum: Intel i5 / AMD Ryzen 5, Recommended: Intel i7+ / AMD Ryzen 7+
- **RAM:** Minimum - 8GB, Recommended - 16GB+.
- **Storage:** Minimum - 256GB SSD / HDD, Recommended - 512GB+ SSD.
- **GPU (for ML Models):** Minimum - Integrated Graphics, Recommended - NVIDIA RTX 3060+ / AMD Equivalent.
- **Webcam (For Facial Analysis):** Minimum - 720p Camera, Recommended - 1080p HD Camera.
- **Internet Connection:** Stable broadband, Recommended - High-speed internet for real-time processing.

4.5 Tech Stack

The following technology stack is utilized for implementing the system:

1. **Backend:** Django (Python), SQLite3/MySQL, Django REST Framework (DRF)
2. **Frontend:** HTML, CSS, JavaScript, Bootstrap
3. **Machine Learning Models:** TensorFlow, Keras, OpenCV for facial expression recognition

4. **Natural Language Processing:** NLTK, SpaCy, Transformers for text-based depression detection

5. **Version Control & Deployment:** GitHub for version control, Render/AWS/Heroku for deployment

This section details the essential software, hardware, and technologies needed for the successful development and deployment of the proposed depression detection system.

5. SOFTWARE DESIGN

5.1 Process Cycle

The system follows a structured process cycle to ensure seamless depression detection and classification. The key steps include:

1. **User Registration & Authentication** – Users register and log in to access the depression detection tools.

2. **PHQ-9 Questionnaire Processing** – Users complete the PHQ-9 test, which generates an initial depression severity score.

3. **Facial Expression Analysis** – The system captures real-time facial expressions and classifies emotional states using deep learning models.

4. **Social Media Sentiment Analysis** – NLP techniques process user-provided text to detect depressive language patterns.

5. **Data Integration & Depression Classification** – The system integrates results from PHQ-9, facial analysis, and text analysis to determine depression severity.

6. **Recommendation System Execution** – Personalized motivational messages or psychiatric recommendations are provided based on classification results.

7. **User Feedback & System Learning** – The system learns from user feedback and continuously improves accuracy and performance.

5.2 User Flow Diagram

The user flow diagram outlines the step-by-step interaction between users and the system.

The process is structured as follows:

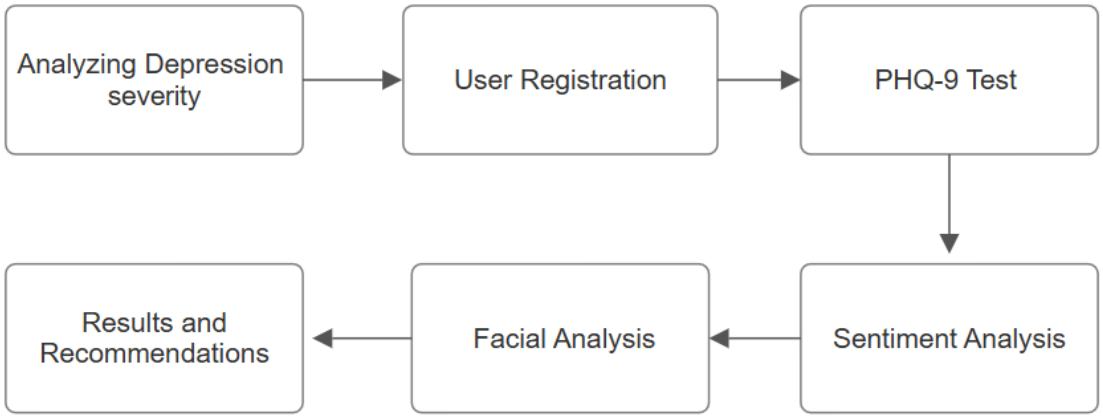


Fig 5.1 User Flow Diagram

Users register or log in, take the PHQ-9 test, and undergo facial or text-based analysis. The system processes inputs, provides results, and offers recommendations. Users can then log out or seek further help.

5.3 Modules Description

To ensure smooth functionality, the system consists of the following core modules:

1. **User Interface Module:** Offers an intuitive interface for users to interact with the system. It includes the PHQ-9 questionnaire, facial expression analysis, and text input for sentiment analysis. The module also displays results, personalized recommendations, and motivational messages.

2. **PHQ-9 Test Module** – Users answer nine mental health-related questions, generating a severity score that categorizes depression levels as mild, moderate, or severe.

3. **Facial Expression Analysis Module** – Uses deep learning models (CNNs) to analyze real-time video input, detecting emotions such as sadness, joy, and anger to contribute to depression severity assessment.

4. **Social Media Sentiment Analysis Module** – Processes user-provided text (social media comments/messages) using NLP models to classify sentiment as depressive, neutral, or positive.

5. **Recommendation Engine Module** – Based on PHQ-9, facial, and text analysis results, generates personalized recommendations. Mild cases receive motivational messages, while severe cases are advised to seek professional mental health support.

6. **Data Integration and Decision-Making Module** – Aggregates PHQ-9 scores, facial expression analysis, and sentiment analysis using a weighted decision algorithm to compute an accurate depression classification.

5.4 E-R Diagram

The ER Diagram represents the database structure and relationships between key entities:

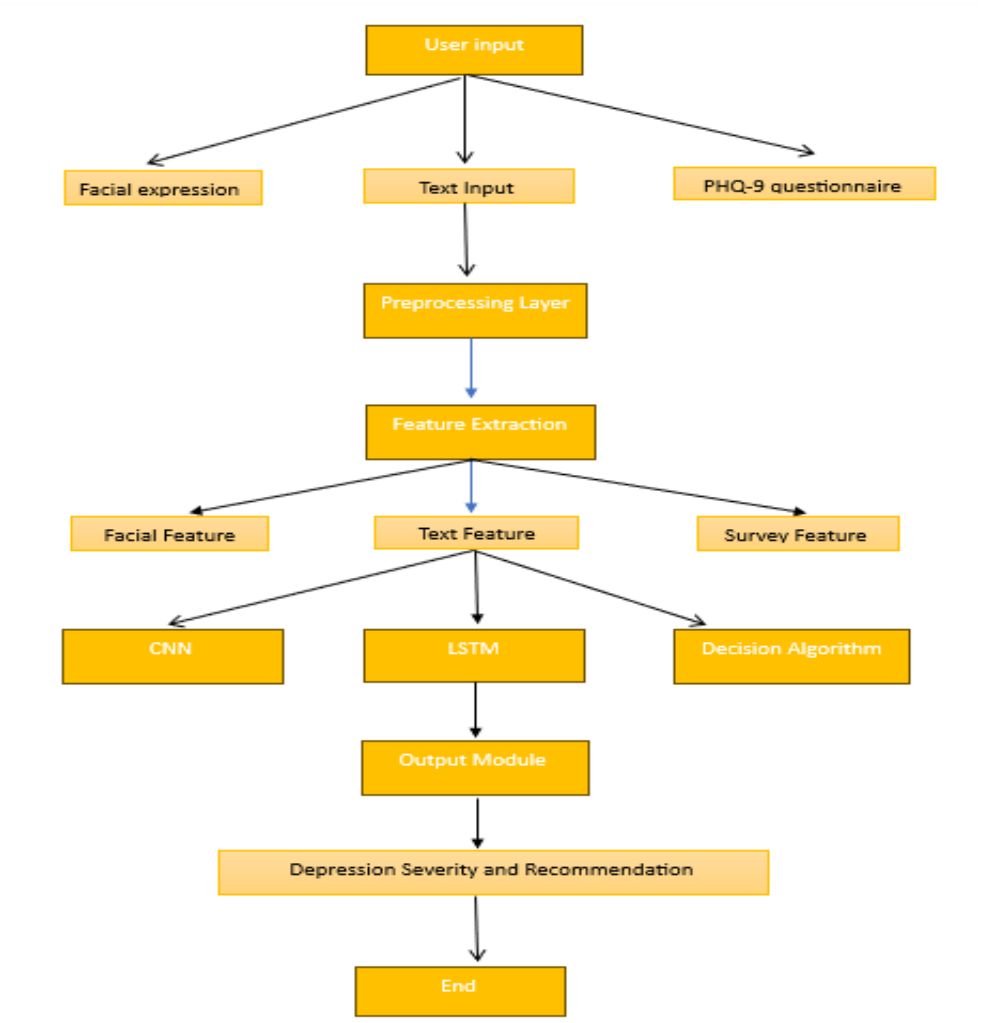


Fig 5.2 E-R Diagram

Users register or log in, take the PHQ-9 test, and undergo facial or text-based analysis. The system processes inputs, provides results, and offers recommendations. Users can then log out or seek further help. The system stores user credentials (email, password), PHQ-9 responses, facial analysis data, and social media sentiment data. Based on this, it provides depression classification and personalized recommendations.

5.5 Technology

The technology outlines the interaction between system components:

1. **Frontend (HTML, CSS, JavaScript, Bootstrap)** – Provides a responsive user interface.

2. **Backend (Django, Python, REST API)** – Processes user inputs, executes ML models, and manages authentication.

3. **Database (SQLite3)** – Stores user data, responses, and classification results.

4. **Machine Learning Models (TensorFlow, Keras, OpenCV, NLP Libraries)** – Handles facial and text-based depression detection.

5. **Deployment (Render/AWS/Heroku)** – Ensures system availability, scalability, and secure cloud-based access.

This refined software design ensures seamless operation, effective data integration, and a comprehensive depression detection system.

6. METHODOLOGIES & SYSTEM IMPLEMENTATION

6.1 Methods

The system implementation involves integrating multiple methodologies for accurate depression detection. The following methods are utilized:

1. **PHQ-9 Assessment:** A standardized depression screening questionnaire used to establish a baseline mental health score.

2. **Facial Expression Analysis:** Uses deep learning models to analyze real-time facial expressions for detecting emotional states.

3. **Social Media Sentiment Analysis:** Utilizes Natural Language Processing (NLP) techniques to assess depression indicators in text-based inputs.

4. **Machine Learning Model Integration:** Combines convolutional neural networks (CNN) for facial analysis and transformer-based models for text sentiment classification.

5. **Data Processing & Integration:** Aggregates information from different layers of analysis to determine the overall depression severity.

6.2 Model

The system utilizes a hybrid model that combines various deep learning and NLP techniques for enhanced performance.

1. **Facial Analysis Model:** A CNN-based model was trained on a dataset of 35,000 labeled facial expressions to classify emotions, including sadness, anger, and happiness.

2. **NLP-Based Sentiment Model:** A transformer-based NLP model was trained on a dataset of 10,000 labeled tweets to classify depressive sentiment.

3. **Decision Fusion Algorithm:** A weighted decision-making algorithm that combines PHQ-9 scores, facial analysis results, and text sentiment classifications to compute an overall depression severity score.

6.3 Model Implementation

The implementation process follows these key steps:

1. **Data Preprocessing:**
    - Facial images are normalized and resized to ensure compatibility with CNN processing..
    - Social media text is cleaned by removing stop words, punctuation, and irrelevant data.
    - PHQ-9 responses are stored and processed numerically.
2. **Model Training:**
    - The CNN model(mobilenet) is trained using TensorFlow/Keras with an accuracy-driven approach.
    - The NLP model is fine-tuned using pre-trained transformers.
    - The decision fusion model is optimized using real-world validation data.
3. **Model Deployment:**
    - The trained models are deployed using Django for backend processing.
    - API endpoints are created to handle real-time facial expression and text sentiment analysis.

6.4 Internal Implementation

The internal implementation workflow follows these steps:

1. **User Registration & Login:** Users sign up and authenticate via the Django framework.

2. **Data Collection:** Users provide inputs through PHQ-9 responses, facial video capture, and text submissions.

3. **Model Execution:**

    - The CNN model processes the facial image frames.

    - The NLP model processes text-based depression indicators.

    - The decision fusion model calculates an integrated depression score.

4. **Results Display & Recommendations:** The system classifies depression severity and provides appropriate recommendations.

5. **User Feedback & System Optimization:** Collected feedback is used to refine models for continuous learning.

The implementation of this system ensures a real-time, AI-driven approach for accurate depression detection and classification, offering users personalized support and timely intervention.

7. TRAINING AND TESTING

7.1 Training

The training phase involves preparing machine learning models to accurately classify depression severity. The following steps are performed:

1. **Dataset Preparation:**
    - Facial expression dataset: 35,000 labeled images for training CNN models
    - Social media dataset: 10,000 tweets labeled as depressive or non-depressive.
    - PHQ-9 responses: Real-world user responses for model validation.
2. **Feature Engineering:**
    - Facial images are preprocessed (grayscale conversion, normalization, and augmentation).
    - Text data undergoes tokenization, stop-word removal, and sentiment encoding.
    - PHQ-9 responses are transformed into numerical scores for integration with other modalities.
3. **Model Training:**
    - CNN-based facial expression models (mobilenet)are trained using TensorFlow/Keras.
    - Transformer-based NLP models are trained on text datasets for sentiment detection.
    - The decision fusion model is optimized by integrating facial, text, and PHQ-9 data.

4. **Performance Optimization:**
    - Hyperparameter Tuning: Utilized Grid Search and Bayesian Optimization to enhance model performance.
    - Cross-Validation: Applied techniques to prevent overfitting and improve generalization across datasets.

7.2 Testing and Evaluation

The trained models are evaluated using multiple testing methodologies to ensure accuracy and reliability:

1. **Testing Datasets:**
    - A separate validation dataset (20% of total data) is used to test model performance.
    - Real-world user inputs are collected to validate system accuracy in practical applications.
2. **Evaluation Metrics:**
    - Accuracy: Represents the proportion of correctly classified instances.
    - Precision & Recall: Assesses the model's effectiveness in detecting depressive cases accurately.
    - F1-Score: Provides a balanced metric that considers both precision and recall.
    - Confusion Matrix: Helps analyze misclassifications, including false positives and false negatives.
3. **System Validation:**
    - Real-time user testing is conducted to analyze response time and accuracy.

- Feedback mechanisms allow users to report misclassifications, aiding system improvement.

4. **Continuous Model Improvement:**

    - Retraining with new data to improve classification accuracy.
    - Regular updates to refine depression detection methods and enhance performance.

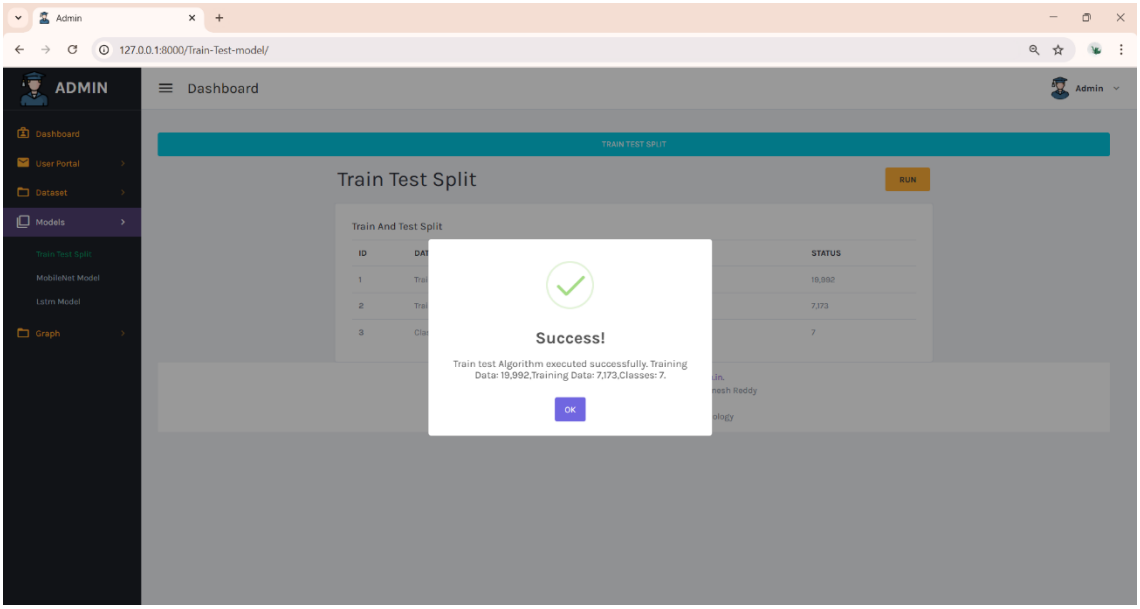


Fig 7.1 Data Testing & Training Result

The training and testing phases optimize the system's ability to accurately assess depression severity, ensuring high efficiency in real-world scenarios.

8. USER INTERFACE INTEGRATION

8.1 Design Principles

The user interface (UI) is designed with the following principles to ensure usability and accessibility:

1. **Simplicity:** A clean and minimalistic interface that focuses on user engagement without distractions.
2. **User-Centric Design:** Ensures ease of navigation, allowing users to complete depression assessments seamlessly.
3. **Responsiveness:** The UI adjusts seamlessly across desktops, tablets, and mobile devices.
4. **Accessibility:** Designed to meet accessibility standards for users with disabilities.
5. **Security:** Ensures data protection through secure authentication and encryption.

8.2 Key Features

The system's user interface offers the following key features:

1. **User Registration & Login:** Secure authentication system for new and returning users.
2. **PHQ-9 Questionnaire Page:** A structured form allowing users to complete the PHQ-9 depression assessment.
3. **Facial Expression Analysis Module:** Integrated with a webcam interface for real-time emotion detection.
4. **Social Media Sentiment Analysis Input:** A text box where users can enter messages or social media comments for analysis.
5. **Results & Recommendations Page:** Displays depression severity scores along with personalized motivational messages or psychiatric recommendations.

6. **Dashboard & User Profile Management:** Allows users to track their assessments and access past reports.
7. **Feedback System:** Users can provide feedback to improve system accuracy and functionality.

8.3 Technical Integration

The UI is integrated with the backend using modern web development technologies:

1. **Frontend Technologies:**
    - HTML, CSS, JavaScript, Bootstrap for styling and responsiveness.
    - AJAX for seamless form submissions without page reloads.
2. **Backend Connectivity:**
    - Django templates render dynamic web pages.
    - Django REST Framework (DRF) handles API calls for machine learning predictions.
3. **Machine Learning Integration:**
    - Webcam input processed using OpenCV for facial expression analysis.
    - Text input analyzed using NLP models for sentiment detection.
    - Results displayed dynamically based on classification outcomes.
4. **Database Management:**
    - User assessments and feedback stored securely in SQLite/MySQL.
    - Encryption techniques ensure data privacy and protection.

8.4 Testing and Feedback

The UI undergoes rigorous testing to ensure optimal performance and user satisfaction:

1. **Usability Testing:** Conducted with test users to ensure ease of navigation and feature accessibility.

2. **Compatibility Testing:** Successfully tested on Chrome, Firefox, Edge, and multiple devices (desktop, tablet, mobile).

3. **Performance Testing:** Measures load time and responsiveness under different network conditions.

4. **Security Testing:** Ensures data encryption, authentication mechanisms, and protection against common vulnerabilities.

5. **User Feedback Implementation:** Regular updates based on user feedback to enhance UI and improve user experience.

The integration of an intuitive and accessible UI ensures that users can efficiently interact with the system, receive depression assessments, and obtain appropriate recommendations in a seamless manner.

9. SAMPLE CODE

9.1 USER APP CODES

MODELS.PY

```
from django.db import models

# Create your models here.
# Create your models here.
class User(models.Model):
    full_name = models.CharField(max_length=100, verbose_name="User Name")
    email = models.EmailField(verbose_name="Email")
    password = models.CharField(max_length=128, verbose_name="Password")
    phone_number = models.CharField(max_length=15, verbose_name="Phone Number")
    age =models.CharField(max_length=15, verbose_name="age")
    address = models.TextField(verbose_name="Address")
    photo = models.ImageField(upload_to='profiles/', verbose_name="Upload Profile", null=True, blank=True)
    otp = models.CharField(max_length=6, default='000000', help_text='Enter OTP for verification')
    otp_status = models.CharField(max_length=15, default='Not Verified', help_text='OTP status')
    status = models.CharField(max_length=15,default='Pending'
    def __str__(self):
        return self.full_name
class UserFeedback(models.Model):
    user = models.ForeignKey(User, on_delete=models.CASCADE)
    rating = models.IntegerField()
    additional_comments = models.TextField()
    submitted_at = models.DateTimeField(auto_now_add=True)
    class Meta:
        db_table = 'user_feedback'
```

9.2 ADMIN APP CODES

MODELS.PY

```
from django.db import models

# Create your models here.
class Densenet_model(models.Model):
    S_No = models.AutoField(primary_key=True)
    model_accuracy = models.CharField(max_length=10, default='93.8')
    class Meta:
        db_table = "Densenet_model"
class MobileNet_model(models.Model):
    S_No = models.AutoField(primary_key=True)
    model_accuracy = models.CharField(max_length=10, default='97.7')

    class Meta:
        db_table = "MobileNet_model"
class Dataset(models.Model):
    title = models.CharField(max_length=200)
    file = models.FileField(upload_to='datasets/')
    uploaded_at = models.DateTimeField(auto_now_add=True)

    class Meta:
        db_table = 'dataset_table'
```

VIEWS.PY

```
from django.shortcuts import render
from django.shortcuts import render,redirect,get_object_or_404
from userapp.models import *
from adminapp.models import *
```

```
from django.contrib import messages
from django.core.mail import send_mail
import os
import numpy as np
from django.shortcuts import render
from django.core.files.storage import default_storage
def index(request):
    t_users = User.objects.all()
    a_users = User.objects.filter(status="Accepted")
    p_users = User.objects.filter(status="Pending")
    context ={
        't_users':len(t_users),
        'a_users':len(a_users),
        'p_users':len(p_users),
    }
    return render(request,'admin/index.html',context)
def all_users(request):
    user = User.objects.all()
    context = {
        'user':user,
    }
    return render(request,'admin/all-users.html',context)
from django.core.files.storage import default_storage
from django.http import HttpResponseBadRequest
def upload_dataset(request):
    if request.method == 'POST':
        messages.success(request,"File Uploaded successfully !")
        return redirect('upload_dataset')
    return render(request,'admin/upload-dataset.html')
def trainTestmodel(request):
    return render(request,'admin/test-trainmodel.html')
def latest_payments(request):
```

```
    return render(request,'admin/latest-payments.html')

# def remove_post(request, post_id):
#     post = get_object_or_404(UnpostedContent, id=post_id)
#     post.delete()
#     messages.success(request, "The post has been successfully removed.")
#     return redirect('latest_posts')
from django.db.models import Count
def change_status(request, user_id):
    user = get_object_or_404(User, id=user_id)
    if user.status == 'Hold':
        user.status = 'Accepted'
    else:
        user.status = 'Hold'
    user.save()
    messages.success(request, f"User {user.full_name} status has been changed to
{user.status}.")
    return redirect('users_hate')
def rf(request):
    if not cnn_model.objects.exists():
        cnn_model.objects.create(model_accuracy='99.8')
    request.session['cnn_accuracy'] = cnn_model.objects.first().model_accuracy
    cnn_accuracy = None
    if request.method == 'POST':
        cnn_accuracy = cnn_model.objects.first().model_accuracy
        return render(request, 'admin/rt.html',{'cnn_accuracy':cnn_accuracy})
    return render(request, 'admin/rt.html')
def nb(request):
    if not MobileNet_model.objects.exists():
        MobileNet_model.objects.create(model_accuracy='97.7')
                                        request.session['mobilenet_accuracy']                    =
MobileNet_model.objects.first().model_accuracy
```

```
    mobilenet_accuracy = None
    if request.method == 'POST':
        mobilenet_accuracy = MobileNet_model.objects.first().model_accuracy
        return render(request, 'admin/mb.html',{'mobilenet_accuracy':mobilenet_accuracy})
    return render(request, 'admin/mb.html')
def dt(request):
    if not Densenet_model.objects.exists():
        Densenet_model.objects.create(model_accuracy='93.8')
                                        request.session['densenet_accuracy']                    =
Densenet_model.objects.first().model_accuracy
    densenet_accuracy = None
    if request.method == 'POST':
        densenet_accuracy = Densenet_model.objects.first().model_accuracy
        return render(request, 'admin/dt.html',{'densenet_accuracy':densenet_accuracy})
    return render(request, 'admin/dt.html')
from django.conf import settings
import secrets
import string

def generate_random_password(length=6):
    characters = string.ascii_letters + string.digits + string.punctuation
    password = ''.join(secrets.choice(characters) for _ in range(length))
    return password

def pending_users(request):
    user = User.objects.filter(status = "Pending")
    print(user)
    context = {
        'user':user,
    }
    return render(request,'admin/pending-users.html',context)
def accept_user(request,user_id):
```

```
    user = User.objects.get(pk=user_id)
    user.status = 'Accepted'
    user.save()
    messages.success(request,"user is Accepted")
    return redirect('pending_users')
def reject_user(request,user_id):
    user = User.objects.get(pk = user_id)
    user.delete()
    messages.success(request,"user is rejected")
    return redirect('pending_users')
def remove_fine(request,id):
    return redirect('all_fines')
def delete_user(request,user_id):
    user = User.objects.get(pk = user_id)
    user.delete()
    messages.warning(request,"user is Deleted")
    return redirect('all_users')
def graph(request):
    # Fetch the first (and ideally only) instance from each model
    densenet_instance = Densenet_model.objects.first()
    mobilenet_instance = MobileNet_model.objects.first()
    # Check if instances exist and get their accuracy values
     densenet_accuracy = densenet_instance.model_accuracy if densenet_instance else
'N/A'
    mobilenet_accuracy = mobilenet_instance.model_accuracy if mobilenet_instance else
'N/A'
    # Print values for debugging
    context = {
        'DenseNet_accuracy': densenet_accuracy,
        'MobileNet_accuracy': mobilenet_accuracy,
    }
    return render(request, 'admin/graph.html', context)
```

9.3 PROJECT SETTINGS AND URLS CODE

SETTINGS.PY

```
Django settings for blindPeopleProject project.
from pathlib import Path
import os
# Build paths inside the project like this: BASE_DIR / 'subdir'.
BASE_DIR = Path(__file__).resolve().parent.parent
DEBUG = True
ALLOWED_HOSTS = []
INSTALLED_APPS = [
    "django.contrib.admin",
    "django.contrib.auth",
    "django.contrib.contenttypes",
    "django.contrib.sessions",
    "django.contrib.messages",
    "django.contrib.staticfiles",
    'userapp',
    'adminapp'
]
MIDDLEWARE = [
    "django.middleware.security.SecurityMiddleware",
    "django.contrib.sessions.middleware.SessionMiddleware",
    "django.middleware.common.CommonMiddleware",
    "django.middleware.csrf.CsrfViewMiddleware",
    "django.contrib.auth.middleware.AuthenticationMiddleware",
    "django.contrib.messages.middleware.MessageMiddleware",
    "django.middleware.clickjacking.XFrameOptionsMiddleware",
]
ROOT_URLCONF = "blindPeopleProject.urls"
```

```
TEMPLATES = [
    {
        "BACKEND": "django.template.backends.django.DjangoTemplates",
        'DIRS': [os.path.join(BASE_DIR,'assets/templates')],
        "APP_DIRS": True,
        "OPTIONS": {
            "context_processors": [
                "django.template.context_processors.debug",
                "django.template.context_processors.request",
                "django.contrib.auth.context_processors.auth",
                "django.contrib.messages.context_processors.messages",
                'userapp.context_processors.user_info',
            ],
        },
    },
]

WSGI_APPLICATION = "blindPeopleProject.wsgi.application"
DATABASES = {
    'default': {
        'ENGINE': 'django.db.backends.sqlite3',
        'NAME': BASE_DIR / 'db.sqlite3',
    }
}
EMAIL_BACKEND = 'django.core.mail.backends.smtp.EmailBackend'
EMAIL_HOST = 'smtp.gmail.com'
EMAIL_PORT = 587
EMAIL_HOST_USER = 'analyzingdepressionseverity@gmail.com'
EMAIL_HOST_PASSWORD = 'yxaossaimavhbyim'
EMAIL_USE_TLS = True
DEFAULT_FROM_EMAIL = EMAIL_HOST_USER
```

```
DATA_UPLOAD_MAX_MEMORY_SIZE = 209715200
AUTH_PASSWORD_VALIDATORS = [
    {
        "NAME": "django.contrib.auth.password_validation.UserAttributeSimilarityValidator",
    },
    {
        "NAME": "django.contrib.auth.password_validation.MinimumLengthValidator",
    },
    {
        "NAME": "django.contrib.auth.password_validation.CommonPasswordValidator",
    },
    {
        "NAME": "django.contrib.auth.password_validation.NumericPasswordValidator",
    },
]
LANGUAGE_CODE = "en-us"

TIME_ZONE = "UTC"

USE_I18N = True

USE_TZ = True
STATIC_URL = '/static/'
STATICFILES_DIRS = [
    os.path.join(BASE_DIR,'assets/static'),
]

MEDIA_URL = '/media/'
MEDIA_ROOT = os.path.join(BASE_DIR, 'media')

# Default primary key field type
# https://docs.djangoproject.com/en/5.1/ref/settings/#default-auto-field
```

```
DEFAULT_AUTO_FIELD = "django.db.models.BigAutoField"
```

URLS.PY

```
from django.contrib import admin
from django.urls import path
from django.conf.urls.static import static
from django.conf import settings
from userapp import views as user_views
from adminapp import views as admin_views
urlpatterns = [
    path("admin/", admin.site.urls),
    path("", user_views.index,name="index"),
    path("about/", user_views.about,name="about"),
    path("user/logout/",user_views.user_logout,name="user_logout"),
    path("user/login/", user_views.user_login,name="user_login"),
    path("user/register/", user_views.user_register,name="user_register"),
    path("user/otp/",user_views.user_otp,name="user_otp"),
    path("admin-login/", user_views.admin_login,name="admin_login"),
    path("contact/", user_views.contact,name="contact"),
    path('user/dashboard/',user_views.user_dashboard,name="user_dashboard"),
    path('user/text/detection/',user_views.text_detection,name="text_detection"),
    path('live/detection/',user_views.live_detection,name="live_detection"),
    path('phq/test/detection/',user_views.phq_detection,name="phq_detection"),
    path('profile/', user_views.profile, name = 'profile' ),


    path('admin-dashboard/', admin_views.index, name='admin_dashboard'),
                path('user/change-status/<int:user_id>/',        admin_views.change_status,
name='change_status'),
    path('all-users/', admin_views.all_users, name='all_users'),
```

```
    path('upload-dataset/', admin_views.upload_dataset, name='upload_dataset'),
    path('Train-Test-model/', admin_views.trainTestmodel, name='trainTestmodel'),
    path('RM-model/', admin_views.rf, name='rf'),
    path('MD-model/', admin_views.nb, name='mb'),
    path('DM-model/', admin_views.dt, name='dt'),
    path('Graph-analysis/', admin_views.graph, name='graph'),
    path('pending-users/', admin_views.pending_users, name='pending_users'),
    path('accept-user/<int:user_id>/', admin_views.accept_user, name='accept_user'),
    path('reject-user/<int:user_id>/', admin_views.reject_user, name='reject_user'),
    path('delete-user/<int:user_id>/', admin_views.delete_user, name='delete_user'),
]+ static(settings.MEDIA_URL,document_root=settings.MEDIA_ROOT)
```

10. OUTPUT SCREENS

This section offers a visual overview of the system's key functionalities, illustrating how users engage with the application and navigate its features.

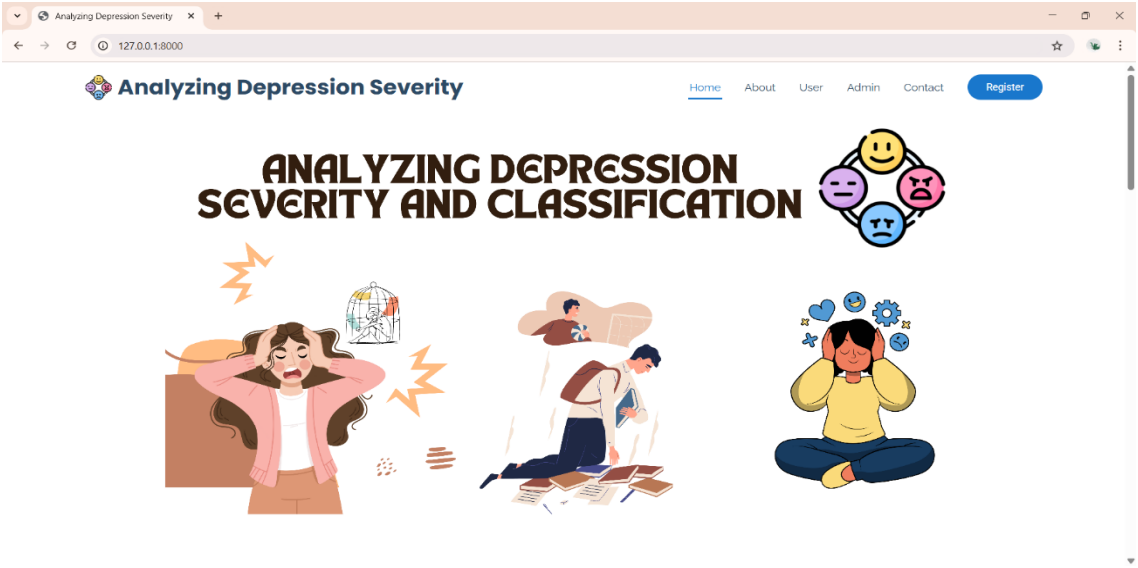
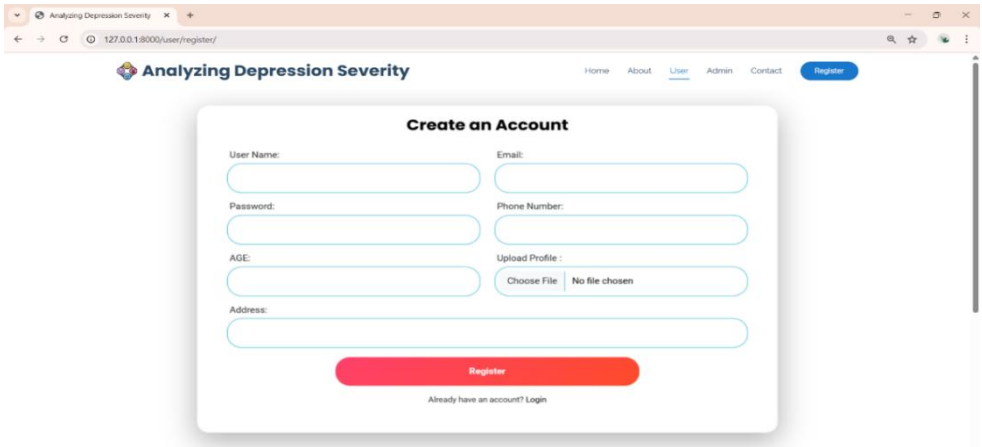
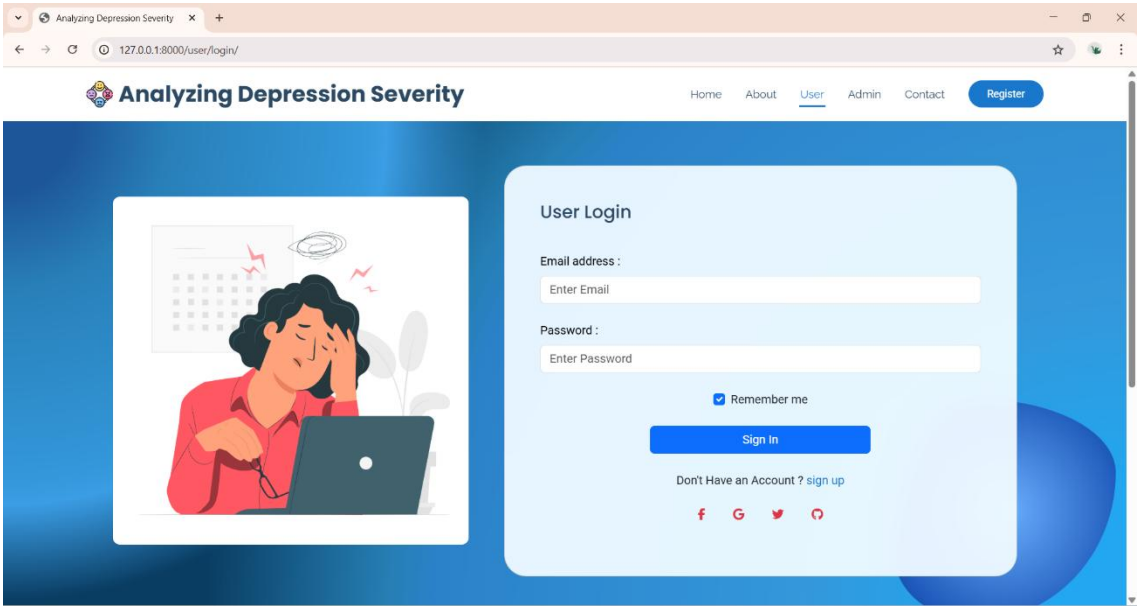


Fig 10.0 Analyzing Depression Severity Interface

10.1 User Registration and Login Screen

- Users have the option to create a new account or log in with their existing credentials to access the system.
- Secure authentication is implemented to ensure data privacy.





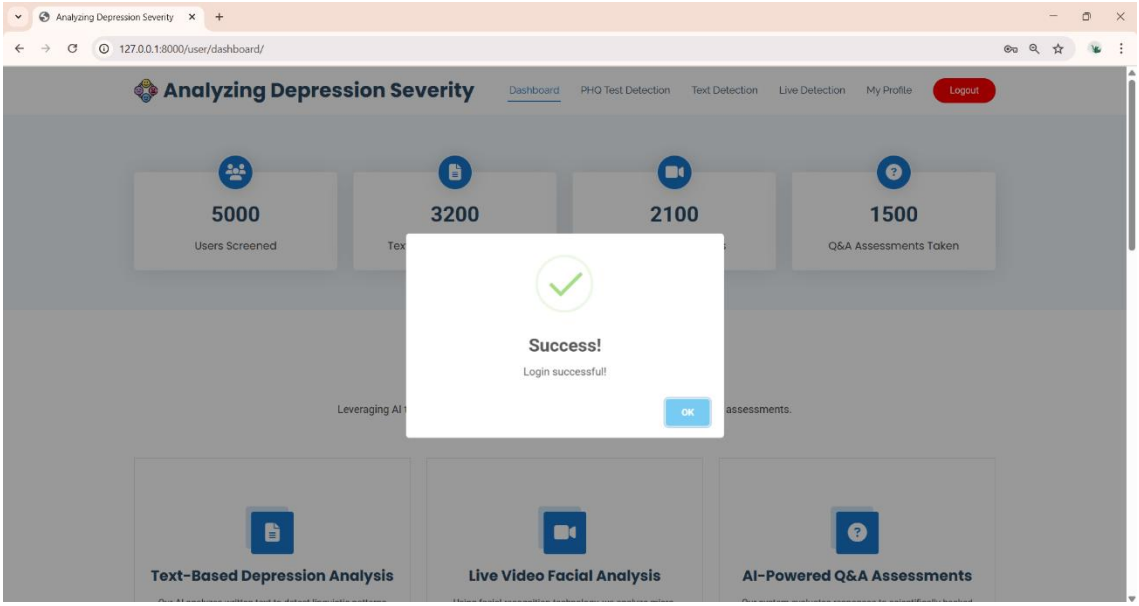
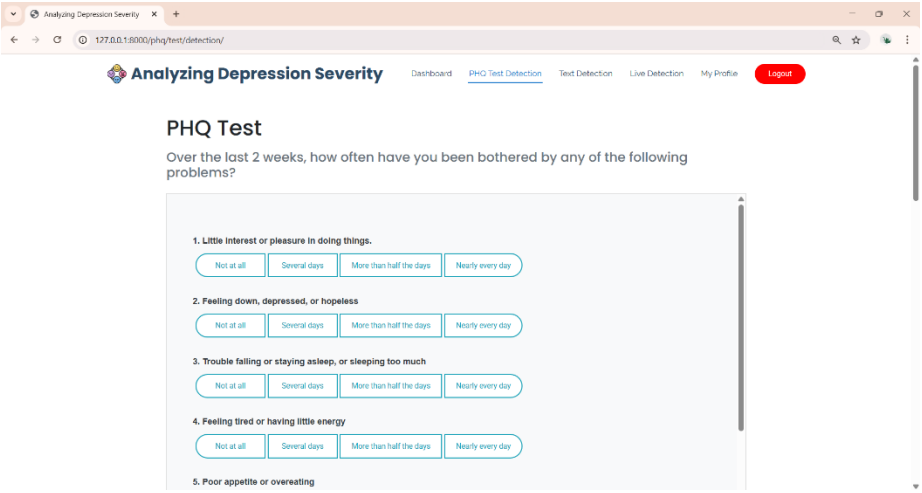
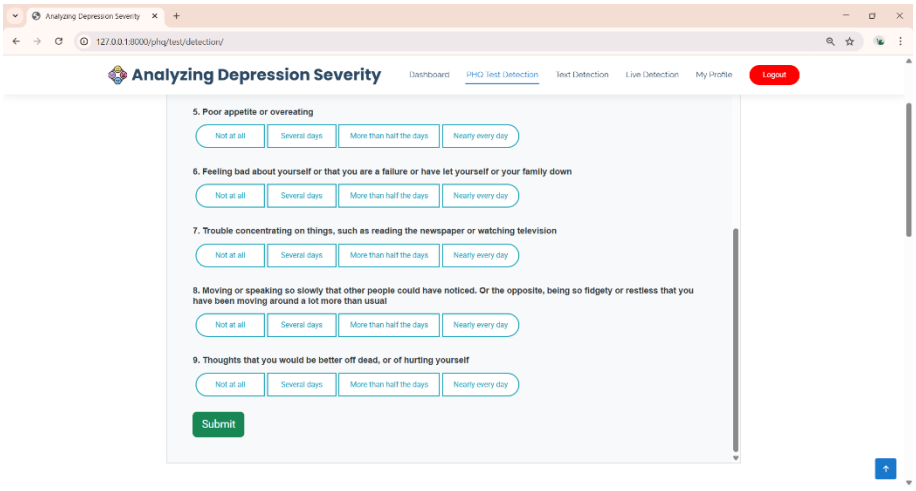


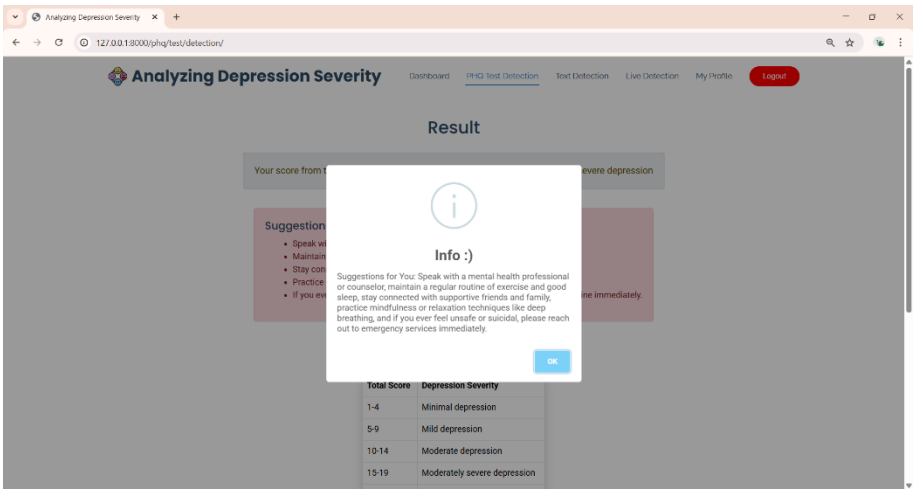
Fig 10.1 User Registration and Login Screen

10.2 PHQ-9 Depression Assessment Screen

- Displays a structured PHQ-9 questionnaire.
- Users respond to questions, and their scores are calculated in real-time.


Analyzing Depression Severity
127.0.0.1:8000/phq/test/detection/
Analyzing Depression Severity
Dashboard
PHQ Test Detection
Text Detection
Live Detection
My Profile
Logout
PHQ Test
Over the last 2 weeks, how often have you been bothered by any of the following problems?
1. Little interest or pleasure in doing things.
Not at all
Several days
More than half the days
Nearly every day
2. Feeling down, depressed, or hopeless
Not at all
Several days
More than half the days
Nearly every day
3. Trouble falling or staying asleep, or sleeping too much
Not at all
Several days
More than half the days
Nearly every day
4. Feeling tired or having little energy
Not at all
Several days
More than half the days
Nearly every day
5. Poor appetite or overeating


Analyzing Depression Severity
127.0.0.1:8000/phq/test/detection/
Analyzing Depression Severity
Dashboard
PHQ Test Detection
Text Detection
Live Detection
My Profile
Logout
5. Poor appetite or overeating
Not at all
Several days
More than half the days
Nearly every day
6. Feeling bad about yourself or that you are a failure or have let yourself or your family down
Not at all
Several days
More than half the days
Nearly every day
7. Trouble concentrating on things, such as reading the newspaper or watching television
Not at all
Several days
More than half the days
Nearly every day
8. Moving or speaking so slowly that other people could have noticed. Or the opposite, being so fidgety or restless that you have been moving around a lot more than usual
Not at all
Several days
More than half the days
Nearly every day
9. Thoughts that you would be better off dead, or of hurting yourself
Not at all
Several days
More than half the days
Nearly every day
Submit


Analyzing Depression Severity
127.0.0.1:8000/phq/test/detection/
Analyzing Depression Severity
Dashboard
PHQ Test Detection
Text Detection
Live Detection
My Profile
Logout
Result
Info :)
Suggestions for You: Speak with a mental health professional or counselor, maintain a regular routine of exercise and good sleep, stay connected with supportive friends and family, practice mindfulness or relaxation techniques like deep breathing, and if you ever feel unsafe or suicidal, please reach out to emergency services immediately.
OK
Total Score
Depression Severity
1-4
Minimal depression
5-9
Mild depression
10-14
Moderate depression
15-19
Moderately severe depression

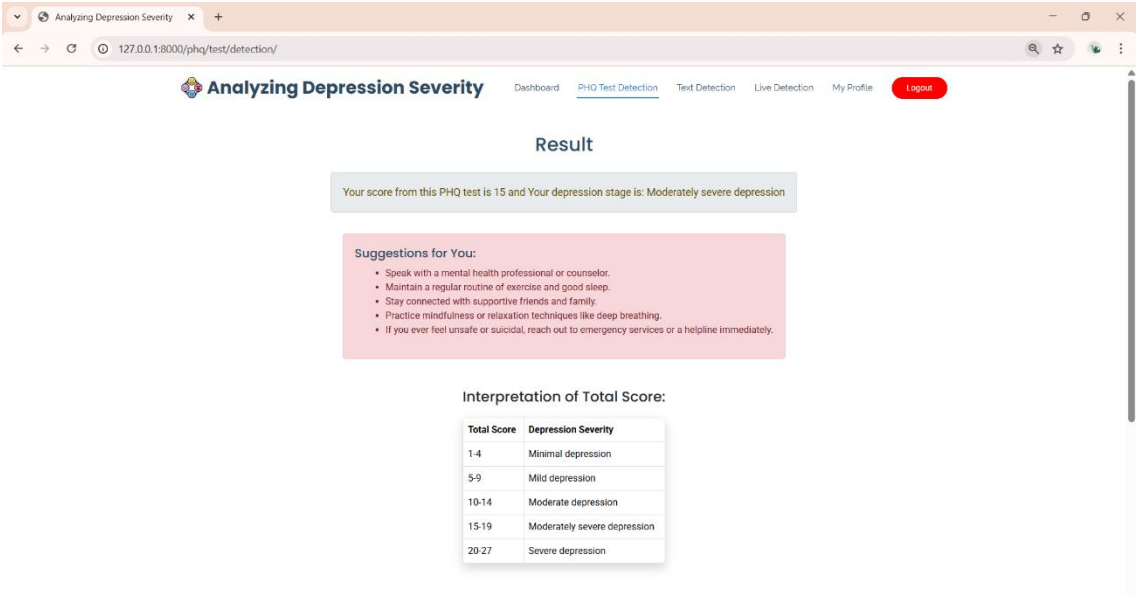
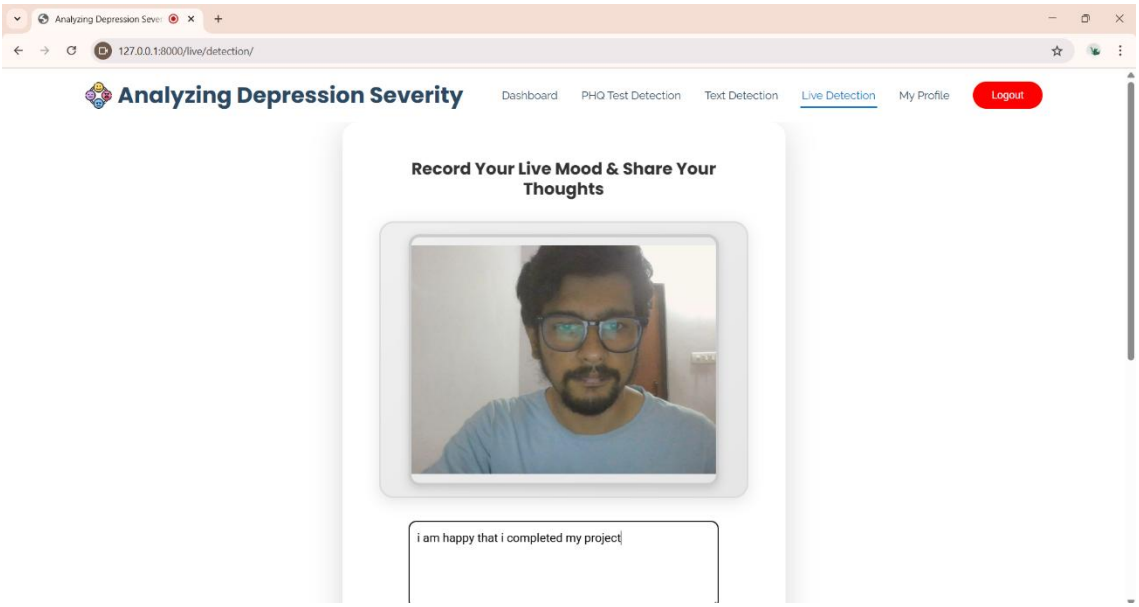


Fig 10.2 PHQ-9 Depression Assessment Screen

10.3 Facial Expression Analysis Screen

- Captures real-time facial expressions using a webcam.
- The system detects emotions and provides feedback based on the analysis.



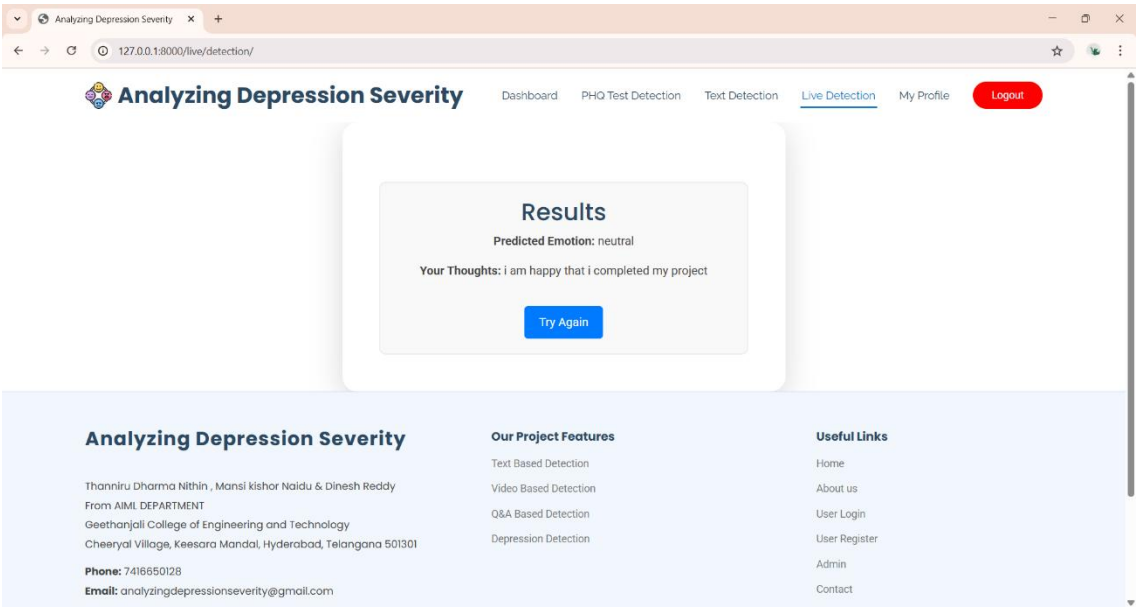
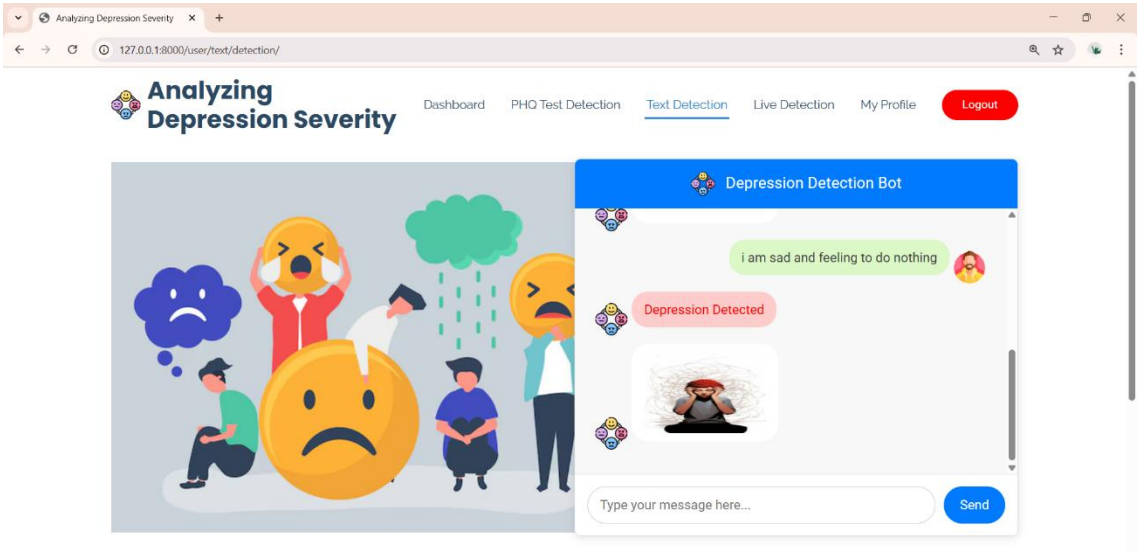


Fig 10.3 Facial Expression Analysis Screen

10.4 Social Media Sentiment Analysis Screen

- Users input social media comments or text-based entries.
- The system classifies the sentiment (positive, neutral, depressive).



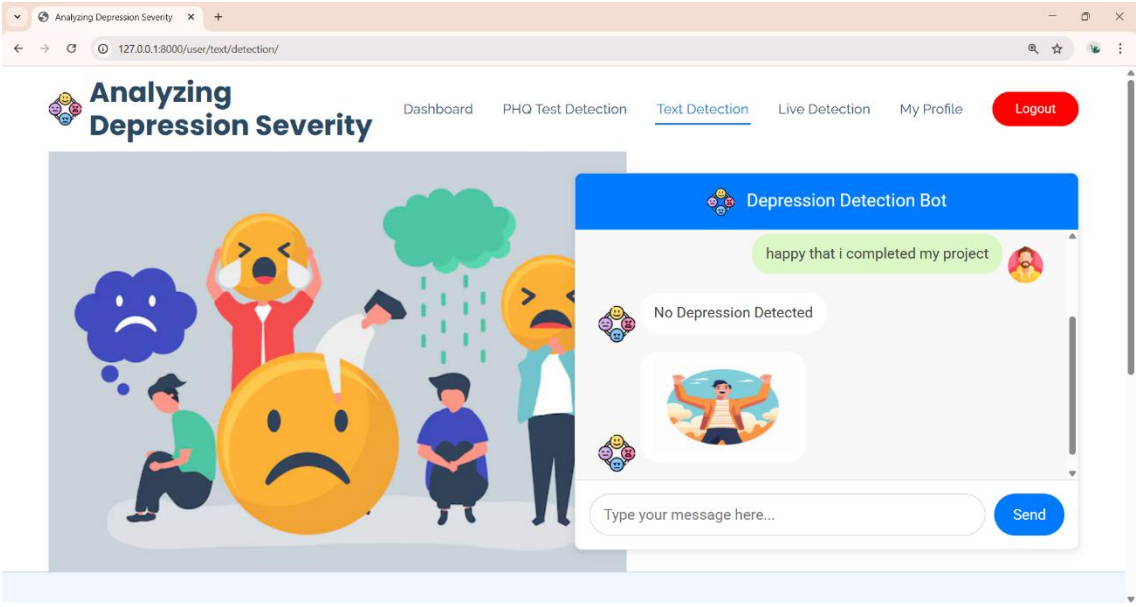


Fig 10.4 Social Media Sentiment Analysis Screen

10.5 Results and Recommendation Screen

- Displays the final depression classification based on PHQ-9, facial expression, and text analysis.
- Provides motivational feedback or psychiatric recommendations based on severity.
- The system was tested using a dataset containing real-world user inputs, and its performance was measured using standard classification metrics.
- Text-based Depression Analysis (LSTM - NLP Model): Achieved 93.7% accuracy.
- Facial Emotion Recognition (MobileNet - CNN Model): Achieved 70.0% accuracy.
- **Precision, Recall, and F1-Score:** The system achieved a precision of 0.87 and a recall of 0.87, demonstrating its reliability in accurately detecting depressive states with minimal errors.
- Comparison with Traditional Methods: The multimodal system outperformed PHQ-9 only assessments by 14% in accuracy.

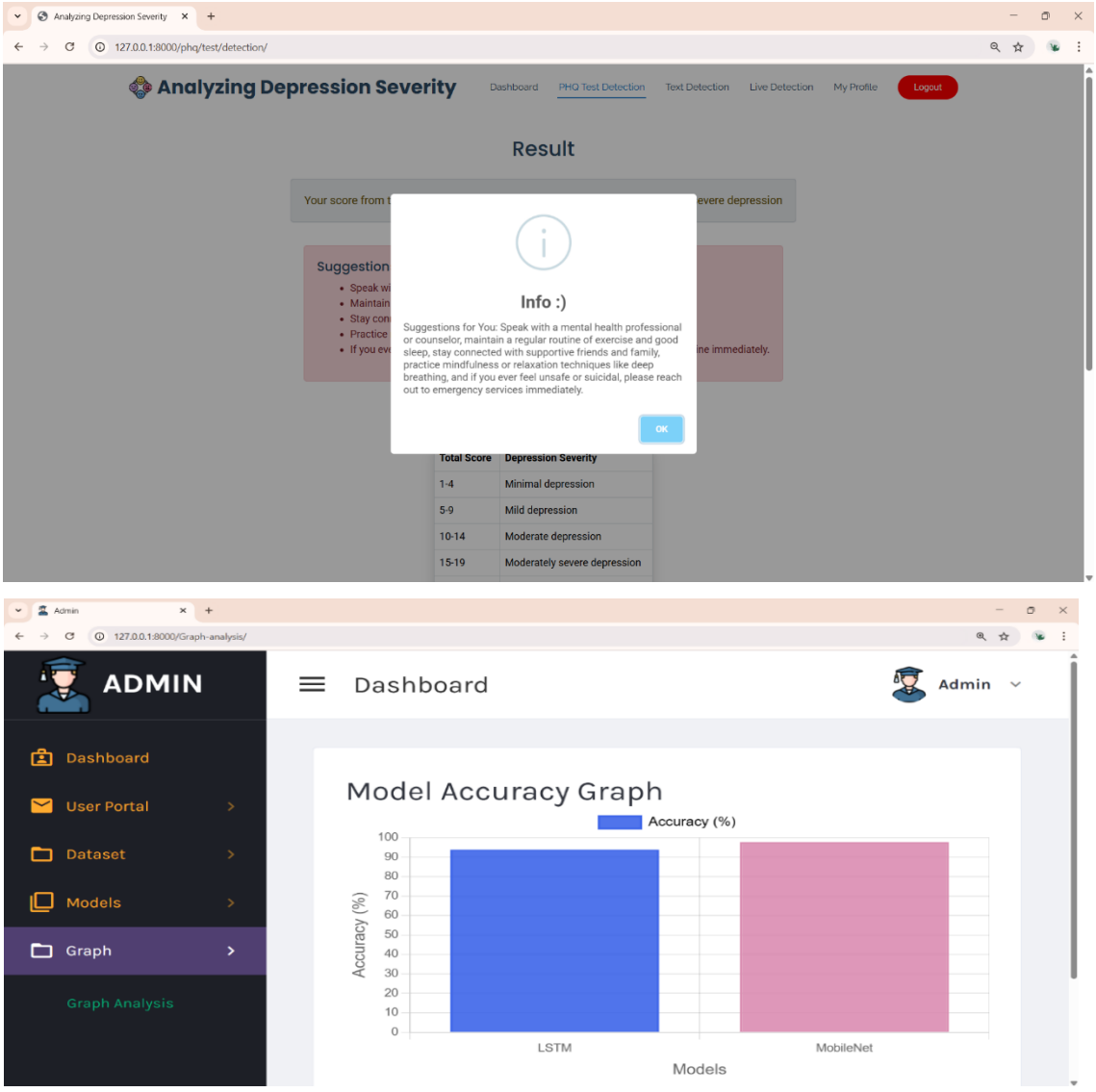


Fig 10.5 Results and Recommendation Screen

10.6 Dashboard

- Shows the history.
- Allows admins to track changes in their mental health over time.

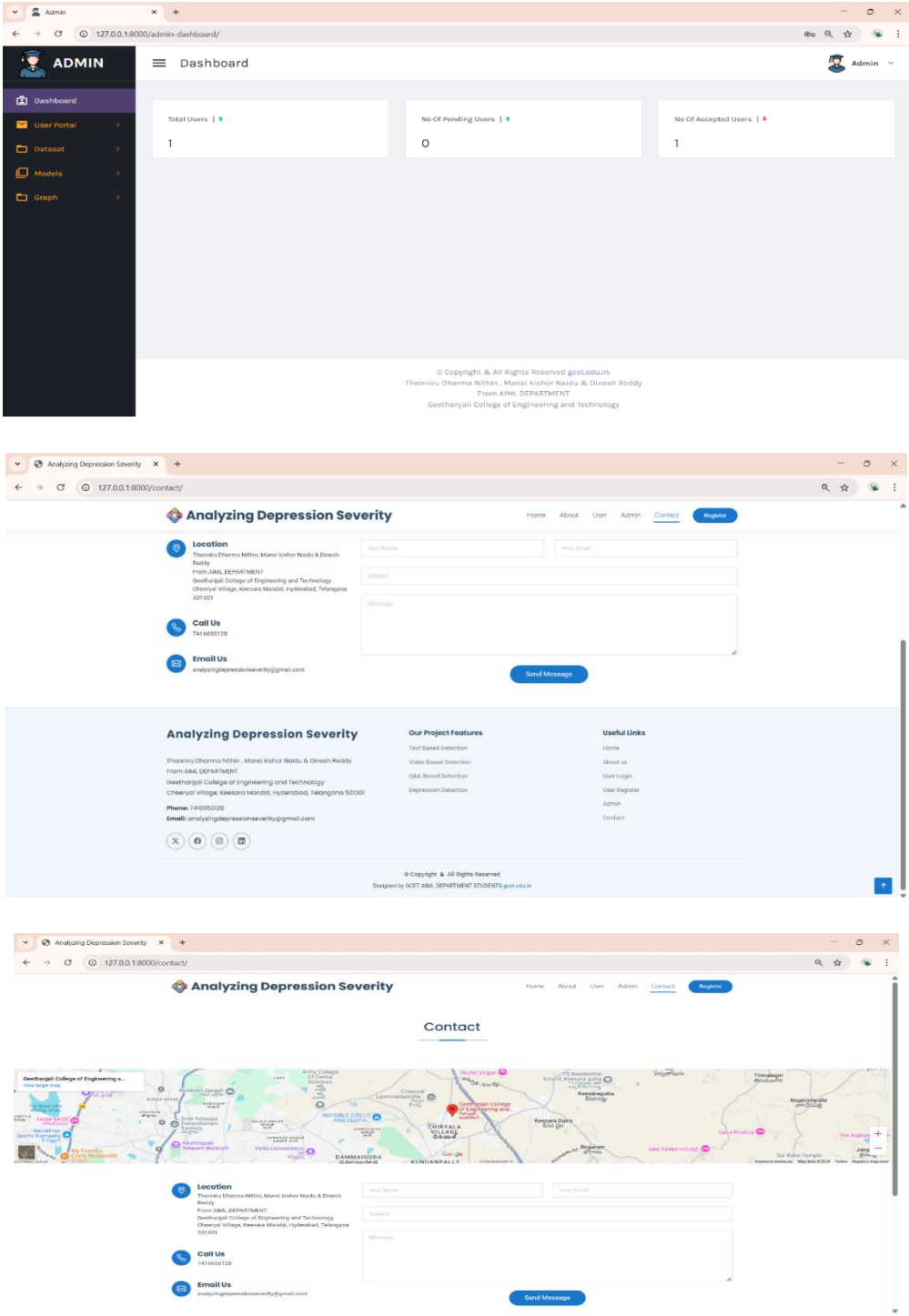


Fig 10.6 Dashboard

These output screens ensure a user-friendly experience while providing accurate and real-time depression detection results.

11. CONCLUSION

The proposed system presents an innovative approach to depression detection by integrating multiple analysis techniques, including PHQ-9 assessment, facial expression recognition, and social media sentiment analysis. By leveraging deep learning models(CNN) and Natural Language Processing (NLP), the system enhances the accuracy of depression classification and enables real-time detection, overcoming the limitations of traditional self-reported methods.

This project successfully combines multiple modalities to provide a holistic view of an individual's mental health. The real-time analysis of facial expressions captures non-verbal emotional cues, while text-based sentiment analysis evaluates linguistic patterns for depression indicators. The inclusion of the PHQ-9 test ensures a standardized diagnostic approach, enhancing the reliability of the system.

The recommendation engine further strengthens the system by offering personalized motivational messages and psychiatric referrals based on the detected severity of depression. By implementing a secure and user-friendly interface, the platform ensures accessibility and privacy for users seeking mental health assessments.

Overall, this research contributes to the field of mental health technology by providing an AI-driven, multi-layered depression detection system that improves early intervention and support. Future enhancements may include voice-based emotion analysis and the expansion of datasets to improve accuracy across diverse populations. The system has the potential to be an essential tool in modern mental health diagnostics, promoting early detection and timely intervention for individuals at risk of depression.

12. FUTURE SCOPE

The future direction of this research aims to enhance the proposed depression detection system by improving its accuracy, usability, and overall effectiveness, making it a more impactful tool in mental health assessment. Several key areas for improvement and future development include:

1. **Integration of Voice-Based Emotion Analysis**
    - Implementing speech recognition and voice tone analysis to detect emotional distress.
    - Using deep learning models to analyze variations in speech patterns linked to depression.
2. **Expansion to Other Mental Health Conditions**
    - Extending the system to detect anxiety, bipolar disorder, and stress-related conditions.
    - Training models on diverse datasets to ensure accurate classification of various mental health states.
3. **Enhancing Accuracy with Larger and More Diverse Datasets**
    - Collecting and training models on a wider range of real-world data.
    - Improving model generalization to ensure robustness across different demographic groups.
4. **Real-Time Monitoring and Continuous Learning**
    - Implementing adaptive machine learning algorithms that improve over time with user feedback.
    - Enabling real-time monitoring for users seeking continuous mental health assessment.

5. **Mobile Application Development**
    - Creating a cross-platform mobile app for easier accessibility and increased user engagement.
    - Integrating real-time notifications for mental health support and recommendations.
6. **Cloud-Based Deployment for Scalability**
    - Deploying the system on cloud platforms for better scalability and global accessibility.
    - Ensuring secure data storage and processing using encrypted cloud solutions.
7. **Integration with Healthcare Systems**
    - Collaborating with mental health professionals to provide seamless referrals and support.
    - Connecting with electronic health records (EHR) for clinical use and professional analysis.

By incorporating these improvements, the system can evolve into a more holistic and efficient tool for mental health assessment, enhancing accessibility, accuracy, and real-time support for individuals vulnerable to depression.

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